Week 7 | Visual design justification

For the Secretary, finance director and journalist: network punctuality improves through a shift toward Metro, while every mode becomes less punctual. Read service first, its decomposition beside it, then corridor use and the fare box. This order prevents an improving headline from being quoted as improved operations.

Use complete 2025 Q2 and 2026 Q2 for service, summing on-time and operated departures before division. The symmetric Kitagawa decomposition separates within-mode decline (about -1.10 points) from mode mix (+2.82). Invalid operational returns are masked and missing days are not zero. This policy slightly changes published anchors that retain impossible rows; the metrics report states the difference. Calendar gaps remain visible and the incomplete September 2026 period is excluded from fare comparisons and interaction.

The corridor map uses annual district-attributed boardings per 1,000 residents, five quantile classes with printed edges, and an equal-area projection. It describes use, not unmet demand or a causal case for construction. Raw boardings would prioritize D17, D09 and D13; rates prioritize D12, D21 and D09. A count-based allocation would take attention away from D12 and D21. D09 combines scale with high use per resident. Count-population correlation is +0.945 versus +0.249 for rates. No district is suppressed or merged (0); ranking should accompany geographic interpretation because 59.9% of people occupy 12.6% of land.

Fare comparisons use journeys, not boardings, in complete April 2025-August 2026 months. Revenue and journeys are indexed separately to 100; they are never added. Revenue grows 26.9% against 12.1% journeys; average yield and mode mix both contribute. Within-mode yield may include ticket-product mix and does not prove a tariff rise. Route district memberships overlap 3.096 times, so no district-sum filter is offered. Hue is constant blue ink, line styles and text identify series, and map lightness encodes only use per resident.

Source: synthetic CDB601220 Week 7; local operator returns, not real-world evidence.